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SORTING AND SEARCHING



There are a few workhorse algorithms we use in nearly every kind of program. Sometimes these algorithms are so fundamental that we take them for granted or don't even realize our code is relying on them.

Several methods for sorting and searching are among these fundamental algorithms. They're worth knowing because they're commonly used and beloved by algorithm enthusiasts (and the sadists who give coding interviews). The implementation of these algorithms can be short and simple, but every character matters, and since they are so commonly needed, computer scientists have striven to enable them to sort and search with mind-melting speed. So we'll also use this chapter to discuss algorithm speed and the special notation we use to compare algorithms' efficiencies.

We start by introducing insertion sort, a simple and intuitive sorting algorithm. We discuss the speed and efficiency of insertion sort and how to measure algorithm efficiency in general. Next, we look at merge sort, a faster algorithm that is the current state of the art for searching. We also explore sleep sort, a strange algorithm that isn't used much in practice but is interesting as a curiosity. Finally, we discuss binary search and show some